

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) Artificial Intelligence (AI) refers to the simulation of human intelligence in machines are programmed to think, learn, and make decisions like humans. Most definitions of AI can be categorised along two key dimensions: 1) human performance versus rationality and 2) thought processes and reasoning (thinking) versus behaviour (action).
- (i) Draw a table with appropriate labels to illustrate the classification of **FOUR (4)** approaches mentioned above. (6 marks)
 - (ii) Define the Turing Test approach and explain how it works. (4 marks)
 - (iii) Identify another name for the Turing Test. (1 mark)
- b) A task environment represents the problems that a rational agent is designed to solve. Therefore, the first step in designing an agent must always be to specify the task environment as fully as possible.
- (i) Describe the **FOUR (4)** components of the **PEAS** framework. (8 marks)
 - (ii) Illustrate how the **PEAS** framework can be applied to describe various task environments for an autonomous delivery drone, as shown in **Figure 1**. Provide **ONE (1)** example for each of the components. (4 marks)



Figure 1. Autonomous Drone for Delivery

- (iii) State whether you agree that the autonomous delivery drone has omniscience. Justify your answer. (2 marks)

[Total: 25 marks]

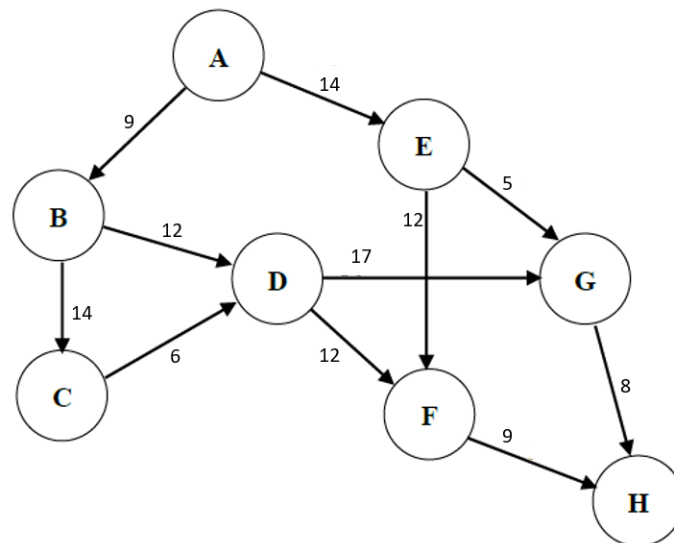
AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 2**

- a) **Hill climbing** is an optimisation algorithm used for solving problems where the goal is to find the best solution by iteratively making small changes. It is commonly used in AI and machine learning for local search optimisation.
- Draw a state-space landscape in which its elevation corresponds to an objective function. (7 marks)
 - Draw a state-space landscape in which its elevation corresponds to a cost function. (7 marks)
 - Describe the **FIVE (5)** labels (current state, global maximum, local maximum, flat local maximum, shoulder) as in **Question 2 a) (i)**. Then, examine the impact of these labels on the efficiency and effectiveness of the hill-climbing algorithm. (10 marks)
 - Identify **ONE (1)** algorithm that can improve the efficiency and effectiveness of the hill-climbing algorithm. (1 mark)

[Total: 25 marks]

Question 3

- a) A software engineering team has developed an AI-powered map navigation application designed to find the shortest path from Station A to Station H. **Figure 4** presents a directed graph consisting of eight stations, where the values within the graph indicate the distance (in kilometers) between each station. **Table 1** displays the heuristic cost of the stations relative to Station H.

**Figure 4.** Search Graph**Table 1:** Heuristic cost of each station

Station	A	B	C	D	E	F	G	H
Euclidean distance	9	8	10	8	7	4	2	1

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3 a) (Continued)**

- (i) Analyse the state space, initial state, goal state, action, transition model, and the cost function for the problem above. (6 marks)
 - (ii) Based on the information provided in **Figure 4** and **Table 1**, illustrate the resulting search tree for A* search in alphabetical order. (8 marks)
 - (iii) Identify the solution returned by A* search in **Question 3 a) (ii)**. (1 mark)
- b) A university needs to schedule classes for four different courses:
1. Mathematics (M)
 2. Computer Science (CS)
 3. Physics (P)
 4. Artificial Intelligence (AI)

The university has three available time slots and three classrooms. Each course must be assigned to a time slot and a room while satisfying several constraints. Mathematics must be scheduled before Computer Science.

Time Slots:

T1 (8:00 AM - 10:00 AM)

T2 (10:00 AM - 12:00 PM)

T3 (2:00 PM - 4:00 PM)

Classrooms:

R1, R2, R3

Lecturer Availability:

Prof. A (M) : Available at T1 or T2

Prof. B (CS) : Available at T2 or T3

Prof. C (P) : Available at T1 or T3

Prof. D (AI) : Available at T2 or T3

- (i) Define a Constraint Satisfaction Problem (CSP). (3 marks)
- (ii) Analyse the case study in **Question 3 b)** as CSP by identifying its variables and each variable's domains. (5 marks)
- (iii) Explain **ONE (1)** real-life example of CSP in the manufacturing field. (2 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4**

- a) A company specialising in automated warehouse management aims to optimise the movement of autonomous robots that transport goods between storage areas and delivery zones. Due to the complexity of the warehouse layout, traditional path-planning algorithms struggle to find the most efficient routes while avoiding congestion. To improve efficiency, the company decides to implement an evolutionary algorithm (EA) to optimise robot navigation. The algorithm evolves potential solutions by simulating different robot movement strategies, evaluating them based on delivery speed, energy consumption, and collision avoidance.
- (i) Analyse how an evolutionary algorithm can help improve the movement of warehouse robots. (2 marks)
 - (ii) Describe **FIVE (5)** phases of an evolutionary algorithm involved in optimising the movement of warehouse robots. (15 marks)
 - (iii) Describe **TWO (2)** stopping criteria for the evolutionary algorithm that ensures the robots achieve optimal performance without excessive computation time. Justify your recommendation. (8 marks)

[Total: 25 marks]